

Database Management Systems

Course Logistics



- **Course name:**
 - Database Management Systems or “DBMS” (CSC-301)
- **Microsoft Team Group:**

Join this group using Team code **rd7en0b**
(Lecture Notes, Tutorials and Assignments etc will be posted here)
- Course-related announcements will be posted on Microsoft Team

Topics to be Covered

- Introduction and data abstraction
- Relational database design, Keys, Database normalization, Functional dependencies, Database decomposition, Normal forms
- Entity-Relationship model
- Relational algebra
- SQL
- Tuple relational calculus
- Transaction management, Serializability, Concurrency control
- Recoverability in DBMS
- Indexing, B and B+ Tree

- Silberschatz, A., Korth, H. F., and Sudershan, S., Database System Concepts, 7th Ed., McGraw-Hill.
 - <https://www.db-book.com/db7/index.html>
- Elmasri, R. and Navathe, S.B., Fundamentals of Database Systems, 7th Ed., Pearson Education.
 - <https://www.pearson.com/us/higher-education/program/Elmasri-Fundamentals-of-Database-Systems-7th-Edition/PGM189052.html>
- Ramakrishnan, R. and Gekhre, J., Database Management Systems, 3rd Ed., McGraw-Hill.
 - <http://pages.cs.wisc.edu/~dbbook/index.html>

- Tutorial and Assignments: 20%
(Tutorial and Assignments may include a programming component)
- Quizzes: 10% (Best 5 out of 8 surprise quizzes)
- Mid-term Exam: 25%
- End-term Exam: 45%